Arithmetic-Derivative Pythagorean 7-Tuples: Exact Extension and Homothetic Structure of OEIS A249107

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Abstract

OEIS A249107 lists the integers whose squares, together with those of their first five arithmetic derivatives, sum to a square. Relative to the accepted ten-term prefix, an exact segmented computation adds six consecutive terms: 1058578125, 1072281223, 1774631673, 2539872501, 9873543431, and 30429212500. The exclusion claim covers all 30048948369 integers from 380264132 through 30429212500 and depends on the executed C implementation and its completed segment records. Independent arithmetic reconstructs all sixteen positive witnesses; independent negative checks cover finite windows, not the full added interval. We give a complete integer-scale criterion for each fixed admissible primitive ordered form, with a residue-core proof for rational logarithmic rates. Applying it to the sixteen witnesses yields eleven observed forms: two inherited infinite homothetic classes and nine singleton classes. Three of the six added terms introduce forms absent from the original prefix. Exact finite trajectory checks and an elementary growth lemma show that three of the sixteen derivative trajectories terminate at zero and the other thirteen diverge. One terminating term satisfies the same square identity at every depth at least five under the explicit zero convention. Finally, $30429212500 < a(17) \le 205397184375$; the upper witness is not assigned an index.

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1 Definition, baseline, and contribution

The arithmetic derivative on the nonnegative integers is defined by

$$\mathrm{D}(0) = \mathrm{D}(1) = 0, \qquad \mathrm{D}(n) = \sum_{p^e \| n} e\frac{n}{p} \quad (n \ge 2). \tag{1}$$

Here p is prime and $p^e \parallel n$ means that e is its exact exponent. It satisfies the product rule

$$\mathrm{D}(ab) = a\mathrm{D}(b) + b\mathrm{D}(a) \tag{2}$$

without a coprimality hypothesis. Indeed, prime exponents add in a product, which proves the identity directly from (1); zero factors are covered by $\mathrm{D}(0) = 0$. It is not additive: $\mathrm{D}(2)+\mathrm{D}(3) = 2$, whereas $\mathrm{D}(5) = 1$. Basic properties, logarithmic equations, and higher iterates are treated by Ufnarovski and Åhlander [2] and Kovič [3]. The survey of Tossavainen and coauthors [4] provides further background; our reading used its accepted author manuscript.

Write $\mathrm{D}^0(n) = n$ and $\mathrm{D}^{j+1}(n) = \mathrm{D}(\mathrm{D}^j(n))$; the superscript denotes composition, not a power of the value. For $h \ge 1$, put

$$Q_h(n) = \sum_{j=0}^{h} \left(\mathrm{D}^j(n)\right)^2, \qquad S_h = \{n \ge 2 : Q_h(n) \text{ is an integer square}\}. \tag{3}$$

We study the increasing enumeration $a(1), a(2), \ldots$ of S_5, with offset 1. Its witness is the ordered seven-coordinate tuple

$$W(n) = (n, \mathrm{D}(n), \mathrm{D}^2(n), \mathrm{D}^3(n), \mathrm{D}^4(n), \mathrm{D}^5(n), R), \qquad R^2 = Q_5(n), \quad R > 0. \tag{4}$$

No primitivity, ordering of magnitudes, common-divisor condition, or positivity condition on later derivatives is imposed. All derivative coordinates are nonnegative, and a zero remains zero thereafter.

Paolo P. Lava introduced A249107 on October 21, 2014. The entry credits Ray Chandler with terms 5–10 in extensions dated December 2016 and January 2017 [1]. The accepted baseline used here has ten terms, ending at $a(10) = 380264131$. A direct read of the entry and its b-file on October 21, 2026 still gives that ten-term baseline. We accept its indexing and gap completeness as prior data; we have not repeated a census of the whole domain below it. The Maple call printed in the entry is an algorithm example, not a retained certificate of an additional negative interval.

This work separates three contributions. First, a completed census beyond the accepted endpoint assigns six additional consecutive indices. Second, normalization and a complete fixed-form scale criterion describe exactly the homothetic classes represented by the sixteen witnesses. Third, finite derivative chains provide rigorous trajectory information and depth comparisons for those witnesses. The two specialized seeds used below and their scaling mechanism were already present in the project analyses before the census. The general fixed-form argument follows the author's earlier studies of A230543 and A249105 [5, 6]; it is proved here at depth five, including the integer-positivity restriction on scales. The exact new ranks, three additional observed forms, and the finite trajectory analysis must be distinguished from those inherited tools. No absolute priority claim is made for the specialized seeds or a general classification of all admissible forms.

2 Complete scales of one admissible form

For a positive integer s, its logarithmic derivative is

$$\mathrm{L}(s) = \frac{\mathrm{D}(s)}{s} = \sum_{p^e \| s} \frac{e}{p}, \qquad \mathrm{L}(1) = 0. \tag{5}$$

In particular $\mathrm{L}(st) = \mathrm{L}(s) + \mathrm{L}(t)$, including when s, t have common prime factors. The rational logarithmic equation and its squarefree denominator restriction are classical [2, Section 8]. Kovič's Proposition 2 classifies the integer-rate case [3].

Definition 2.1. For a witness W, let g be the gcd of all seven coordinates, including its root, and let

$$F = W/g = (u_0, u_1, u_2, u_3, u_4, u_5, r), \qquad u_0 = m.$$

This is its primitive ordered form. A nonnegative integer tuple of this shape, with $m, r > 0$ and gcd 1, is admissible if some positive integer multiple is a witness with input at least 2. Its homothetic class consists of all such witnesses sF.

The order of coordinates is retained. A primitive form need not itself be a derivative chain, since dividing a chain by g does not commute with arithmetic differentiation.

Theorem 2.2 (Fixed-form criterion)**.** *Let F be an admissible primitive ordered form and define*

$$\lambda = \frac{u_1 - \mathrm{D}(m)}{m}. \tag{6}$$

Then $\lambda \geq 0$, and

$$u_{j+1} = \mathrm{D}(u_j) + \lambda u_j \quad (0 \leq j < 5), \qquad \sum_{j=0}^{5} u_j^2 = r^2. \tag{7}$$

For every positive integer s with $sm \geq 2$, the tuple sF is a witness if and only if $\mathrm{L}(s) = \lambda$. No noninteger positive rational multiplier makes all seven coordinates of F integral.

Proof. Choose a witnessing scale t. From $\mathrm{D}(tm) = tu_1$ and the product rule, $\mathrm{D}(m) + m\mathrm{L}(t) = u_1$, so $\lambda = \mathrm{L}(t) \geq 0$. Each remaining transition $\mathrm{D}(tu_j) = tu_{j+1}$ gives the corresponding equality in (7). If $u_j = 0$, this equality forces $u_{j+1} = 0$, so the shifted transition also holds at zero. The geometric identity follows by dividing the original square identity by t^2.

For a proposed s, the first derivative equality $\mathrm{D}(sm) = su_1$ is equivalent to $\mathrm{L}(s) = \lambda$. If it holds, induction gives

$$\mathrm{D}(su_j) = s\mathrm{D}(u_j) + u_j\mathrm{D}(s) = s\big(\mathrm{D}(u_j) + \lambda u_j\big) = su_{j+1} \quad (0 \leq j < 5).$$

This remains valid at a zero coordinate. The square identity scales by s^2, proving sufficiency. Finally, if $a/b > 0$ is reduced and $(a/b)F$ is integral, then b divides every coordinate of F, and hence $b = 1$. □

Only the product rule and integer arguments were used. In particular, there is no binomial expansion of a purported additive operator. For an unwitnessed proposed form, the square identity, all five transitions, the scale equation, and the input domain must each be verified.

Theorem 2.3 (Residue core and all positive integer scales)**.** *Let $\lambda = A/B \geq 0$ be reduced, with $B > 0$. If $\mathrm{L}(s) = \lambda$ has a positive integer solution, then B is squarefree. When B is squarefree, define, for every prime $p \mid B$,*

$$c_p \equiv A(B/p)^{-1} \pmod{p}, \qquad 1 \leq c_p < p, \tag{8}$$

and put

$$s_0 = \prod_{p|B} p^{c_p}, \qquad \alpha_0 = \sum_{p|B} \frac{c_p}{p}, \qquad k = \lambda - \alpha_0. \tag{9}$$

For $B = 1$, use $s_0 = 1$ and $\alpha_0 = 0$. Then k is an integer. There is a positive integer solution if and only if $k \geq 0$, in which case all and only the solutions are

$$s = s_0 \prod_{j=1}^{k} p_j^{p_j}, \qquad p_j \in \mathbb{P}. \tag{10}$$

Prime parameters may repeat and may divide B. For an admissible form, the scales in (10) *are exactly its homothetic class.*

Proof. Every $\mathrm{L}(s) = \sum e_p/p$ has reduced denominator dividing the squarefree product of the primes of s. Suppose now that B is squarefree. The residue representatives exist and are nonzero since $\gcd(A, B) = 1$. Furthermore

$$Bk = A - \sum_{p|B} c_p(B/p)$$

is an integer divisible by every prime of B, by (8). Since B is squarefree, B divides this integer, proving $k \in \mathbb{Z}$. For $B = 1$ the assertion is immediate.

For any solution $s = \prod p^{e_p}$, let M be the squarefree product of the primes dividing sB. Multiplication of the rate equation by M gives

$$\sum_{p|s} e_p M/p = AM/B. \tag{11}$$

If $p \mid B$ were absent from s, reduction modulo p would give zero on the left and a nonzero number on the right. Thus $p \mid s$. For such a prime, reduction and cancellation of the unit M/B modulo p

give $e_p(B/p) \equiv A \pmod p$. For a prime of s outside B, reduction instead gives $e_p \equiv 0 \pmod p$. Consequently

$$e_p = c_p + ph_p \ (p \mid B), \qquad e_p = ph_p \ (p \nmid B), \qquad h_p \in \mathbb{Z}_{\geq 0}.$$

Substitution yields $\lambda = \alpha_0 + \sum_p h_p$, so $k \geq 0$ and listing each prime h_p times gives (10). Conversely, every displayed product has rate $\alpha_0 + k = \lambda$. For $k \geq 0$, choosing any k primes proves existence. The fixed-form conclusion is Theorem 2.2; the input restriction causes no exception for an admissible form. The only possible failure of $sm \geq 2$ would be $s = m = 1$. Since the proposed scale satisfies $\mathrm{L}(s) = \lambda$, it would give $\lambda = 0$ and, by (7), $u_1 = \cdots = u_5 = 0$. Any witnessing scale t would then also satisfy $\mathrm{L}(t) = 0$, hence $t = 1$ by (5). Its input would be $tm = 1$, contradicting admissibility. □

Corollary 2.4. *An admissible fixed form has a singleton homothetic class precisely when $k = 0$, and an infinite class when $k \geq 1$. At an integer rate λ, its scales are products of exactly λ factors p^p, with repetition.*

Proof. For $k = 0$ the sole scale is s_0. For $k \geq 1$, fix $k - 1$ prime parameters and let the remaining prime increase. This gives infinitely many distinct scales and first coordinates. For an integer rate, $B = 1$ and $k = \lambda$. □

The denominator condition alone is insufficient in the integer domain. For $\lambda = 1/6$, one has $c_2 = 1$, $c_3 = 2$, $s_0 = 18$, $\alpha_0 = 7/6$, and $k = -1$, so there is no positive integer scale. In contrast, rate $1/2$ has the sole scale 2. This last example concerns the rate equation alone, not an exhibited A249107 form. Thus "singleton" means $k = 0$, not necessarily $\lambda = 0$. These results classify the scales of a fixed form; they do not classify all admissible forms.

3 The inherited families and the eleven observed forms

Two primitive ordered seed tuples present before the campaign are

$$F_1 = (338745, 550369, 550370, 947809, 954003, 1272007, 2036425), \tag{12}$$

$$F_2 = (2434337, 5075913, 11896417, 25623221, 53701605, 153944646, 165567453). \tag{13}$$

Their rates are respectively 1 and 2. In particular, $\mathrm{D}(338745) = 211624$ and $\mathrm{D}(2434337) = 207239$; neither tuple is an unshifted derivative chain. Table 1 gives the five exact derivative evaluations needed to check the shifted transitions. Direct summation of the six squares gives respectively 2036425^2 and 165567453^2; the gcd of each complete seven-tuple is 1.

Form	$\mathrm{D}(u_0)$	$\mathrm{D}(u_1)$	$\mathrm{D}(u_2)$	$\mathrm{D}(u_3)$	$\mathrm{D}(u_4)$
F_1	211624	1	397439	6194	318004
F_2	207239	1744591	1830387	2455163	46541436

Table 1: Exact transition checks for the inherited primitive forms. Adding λu_j, with $\lambda = 1$ or 2 respectively, gives u_{j+1} at each of the five positions.

Proposition 3.1. *The complete homothetic classes of F_1 and F_2 have first coordinates*

$$n = 338745p^p, \qquad n = 2434337p^p q^q, \qquad p, q \in \mathbb{P}, \tag{14}$$

respectively. Both are infinite subsets of A249107. Repeated parameters and primes dividing the seed are allowed.

Proof. Equation (1) gives $\mathrm{D}(p^p) = p^p$. Additivity of L on products gives $\mathrm{L}(p^p q^q) = 2$, also for $p = q$. The checked transitions and square identities make the scaled tuples witnesses by the product-rule induction of Theorem 2.2. Theorem 2.3 proves that these are all their scales. Infinitude follows from Corollary 2.4. □

Form	Indices represented	Observed scales g	(λ, k)
F_1	$5, 6, 11$	$4, 27, 3125$	$(1, 1)$
F_2	$7, 9, 13, 16$	$16, 108, 729, 12500$	$(2, 2)$
T_1	1	1	$(0, 0)$
T_2	2	1	$(0, 0)$
T_3	3	1	$(0, 0)$
T_4	4	1	$(0, 0)$
T_8	8	1	$(0, 0)$
T_{10}	10	1	$(0, 0)$
T_{12}	12	1	$(0, 0)$
T_{14}	14	1	$(0, 0)$
T_{15}	15	1	$(0, 0)$

Table 2: All primitive ordered forms observed in the sixteen-term prefix. The full coordinates of F_1, F_2 are (12)–(13); T_i is exactly the seven-coordinate witness in row i of Table 5. Every residue core s_0 in this table is 1. Only the two displayed classes, rather than all possible infinite classes, are being enumerated.

Proposition 3.2 (Classification of the observed forms)**.** *The sixteen witnesses in Table 5 have exactly the eleven primitive ordered forms in Table 2. The nine T_i have singleton homothetic classes over the entire positive integer scale domain.*

Proof. Take the gcd of all seven coordinates in each row. The nonunit gcds are precisely the seven scales shown for F_1, F_2. The other nine rows have gcd 1 and rate zero, and hence are their own primitive derivative witnesses. Their first coordinates are distinct, and differ from those of F_1, F_2, so the eleven tuples are distinct. All derivative and square identities were reconstructed using exhaustive trial division and integer square roots. For each T_i, $\mathrm{L}(s) = 0$ forces $s = 1$ by (5), proving the global singleton claim for that fixed form. □

The original ten rows represent eight forms, including six singletons. The added indices 12, 14, and 15 introduce three forms absent from that sample; indices 11, 13, and 16 enlarge the represented scales of the inherited families. Specifically, they correspond to $F_1(5)$, $F_2(3, 3)$, and $F_2(2, 5)$. Their positive values were available from the family constructions before the missing intervals were enumerated. The census supplies their exact indices. All nine singleton first coordinates in this sample are squarefree; the other seven first coordinates are not. This is a finite observation, not a general equivalence between squarefreeness and singleton homothetic classes.

4 Derivative trajectories and the effect of changing depth

The square identity concerns a finite initial trajectory. Its later behavior can nevertheless be decided for every term in the present prefix, using only three additional derivatives and the following historical growth mechanism [2, Theorem 4].

Lemma 4.1. *If an iterate of n is $p^p b$ for a prime p and an integer $b > 1$, then $\mathrm{D}^j(n) \longrightarrow \infty$ as $j \longrightarrow \infty$.*

Proof. The product rule gives $\mathrm{D}(p^p b) = p^p(b+\mathrm{D}(b))$. Starting at this iterate, put $b_{t+1} = b_t + \mathrm{D}(b_t)$. Every $b_t > 1$ has a nonempty prime factorization, so (1) gives $\mathrm{D}(b_t) \geq 1$ and $b_t \geq b_0 + t$. Thus the trajectory is unbounded and tends to infinity. No conjecture about arbitrary arithmetic-derivative trajectories is needed. □

i	$\mathrm{D}^6(a(i))$	$\mathrm{D}^7(a(i))$	$\mathrm{D}^8(a(i))$	Eventual behavior
1	12032	48384	248832	diverges; $4 \mid \mathrm{D}(a(1)) = 168$
2	10260	23112	57996	diverges; $4 \mid \mathrm{D}(a(2)) = 216$
3	4579200	22524480	104443776	diverges; $4 \mid \mathrm{D}(a(3)) = 24544$
4	1202048	4207296	14275072	diverges; $4 \mid \mathrm{D}(a(4)) = 7484$
8	1	0	0	first zero at depth 7
10	67928168	117772180	143351236	diverges; $4 \mid \mathrm{D}(a(10)) = 29251100$
12	0	0	0	first zero at depth 6
14	201791	1	0	first zero at depth 8
15	4208230	3269601	2285484	diverges; $4 \mid \mathrm{D}^8(a(15))$

Table 3: Exact continuation of the nine singleton witnesses. Every displayed multiple of 4 used for divergence is greater than 4. The seven represented members of F_1, F_2 already contain a proper factor p^p at depth zero.

Proposition 4.2 (Complete trajectory decision for the finite prefix)**.** *For $i = 8, 12, 14$, the trajectory of $a(i)$ first reaches zero at depths $7, 6, 8$, respectively. For every other $1 \leq i \leq 16$, it tends to infinity.*

Proof. The exact continuations in Table 3, together with the positive coordinates of Table 5, prove the three first-zero claims. The six other singleton rows have a displayed iterate equal to $4b$ with $b > 1$, so Lemma 4.1 applies with $p = 2$. For F_1, every scale p^p is a proper divisor of its first coordinate, since $338745 > 1$. For F_2, one factor p^p leaves the multiplier $2434337q^q > 1$. The lemma therefore applies to every member of both families, including repeated or shared primes. □

This proposition decides sixteen particular trajectories, and in addition the trajectories of all members of the two proved families. It does not settle the general dynamics of D, or prove that no other behavior can occur outside these sets. A singleton homothetic class can thus contain either a terminating or a divergent trajectory.

Proposition 4.3 (A persistent square identity)**.** *Let $N = 1072281223 = a(12)$ and $R = 1076009463$. Then*

$$\mathrm{D}^5(N) = 1, \quad \mathrm{D}^j(N) = 0 \ (j \geq 6), \quad Q_h(N) = R^2 \ (h \geq 5). \tag{15}$$

For the zero convention in (1)*, the sixteen-term sample has no member of S_h for $1 \leq h \leq 4$, and its only member of S_h for each $h = 6, 7, 8$ is N.*

Proof. The witness in row 12 has fifth derivative 1. The definitions give its zero tail, and adding zero squares proves (15) at every later depth. For the finite assertions, reconstruct the sixteen chains through depth 8 and test each partial sum using $t = \lfloor \sqrt{Q_h} \rfloor$ and $t^2 = Q_h$. No input interval outside these sixteen values is searched. □

The other two terminating cases illustrate why termination alone is not enough. For $a(8)$, the stabilized sum from depth 6 onward is $53881926^2 + 1$, strictly between consecutive squares. For $a(14)$ it stabilizes from depth 7 onward at $2705738777^2 + 201791^2 + 1$, also nonsquare by exact integer-root testing. Different square-sum depths are not generally nested: the fixed point $n = 4$ satisfies $Q_3(4) = 64$ and $Q_5(4) = 96$, while $4031 \in S_5$ and $4031 \notin S_3$ by its exact partial sum.

5 Exact enumeration and numerical domain

5.1 Arithmetic kernel and coverage

The production program is a specialized C17 implementation for Q_5, with GCC's unsigned 128-bit extension, OpenMP, GMP, and OpenSSL libcrypto. It enumerates every integer in a configured inclusive interval, using segmentation for arithmetic and recovery rather than a restriction to known families.

For a block $[L, H]$, a prime sieve supplies all primes through $\lfloor\sqrt{H}\rfloor$. At each multiple n of p, divide its residual by p until no longer divisible, and add $e(n/p)$ to the first derivative, where n is the original input. Any residual greater than 1 is prime: a composite residual would have a prime divisor at most $\sqrt{H}$ that the sieve had already removed. Adding n/residual completes $\mathrm{D}(n)$ exactly.

For nonprime inputs, four further factorizations compute $\mathrm{D}^2(n), \ldots, \mathrm{D}^5(n)$. The word factorizer first tests primality using strong probable-prime tests with bases

$$2, 325, 9375, 28178, 450775, 9780504, 1795265022. \tag{16}$$

The deterministic coverage for inputs below 2^{64} is the finite-domain record attributed to Jim Sinclair in the Miller–Rabin catalogue [8]; the machine-word setting is also discussed by Forišek and Jančina [9]. This external finite-domain result is a premise of the computation, not a new exhaustive primality verification in this project. Bases are reduced modulo the tested integer; a zero residue is skipped while the remaining bases are still tested.

Composite operands are split by fixed-seed Pollard-rho iterations, using the factor-finding principle of [10]. The source uses $x \mapsto x^2 + c$ with $1 \le c \le 32$, at most 100000 steps per seed, and gcd comparisons of one-step and two-step trajectories. Failure of the bounded fast path leads to exhaustive odd trial division, after small primes through 97 have been tried. The trial condition $p \le z/p$ avoids a squaring overflow. Every recursive split is a proper divisor, so the operands strictly decrease. At prime leaves the program checks the reconstructed product and accumulates exact multiplicities in (1). The finite fallback and proper descent establish termination in the absence of interruption or resource failure, although they supply no useful worst-case runtime guarantee for every operand.

The only early mathematical rejection is $\mathrm{D}(n) = 1$: this characterizes prime original inputs, for which $Q_5(n) = n^2+1$ lies strictly between n^2 and $(n+1)^2$. Subsequent ones and zeros are evaluated normally. Indeed a composite $n = ab$, with $a, b \ge 2$, has $\mathrm{D}(n) = a\mathrm{D}(b) + b\mathrm{D}(a) \ge a + b \ge 4$. There is no family filter, floating-point square decision, or probable-prime acceptance beyond the stated finite deterministic primality domain. Each of the six coordinates is imported into GMP before squaring and summation. GMP's exact square predicate decides membership; an integer square root is then obtained for a positive result [11].

5.2 Integer bounds

Let $\Omega(z)$ count prime factors with multiplicity. For $z \ge 2$,

$$\mathrm{D}(z) \le z\Omega(z)/2 \le z\lfloor\log_2 z\rfloor/2. \tag{17}$$

The first inequality follows because each reciprocal prime is at most $1/2$; the second follows from $2^{\Omega(z)} \le z$. This is the usual arithmetic-derivative bound [2], in a convenient integer form. The function $U \mapsto \lfloor U\lfloor\log_2 U\rfloor/2\rfloor$ is nondecreasing for integer $U \ge 2$: both nonnegative factors U and $\lfloor\log_2 U\rfloor$ are nondecreasing, as is taking the floor. Iterating it at the supported input cap gives Table 4, with 0 and 1 handled separately.

The sum of the six squared bounds is

$$Q_5(n) \le 3063711203608923812500000000000000000000000,$$

Quantity	Universal upper bound at the source cap	Representation
n	40000000000000	unsigned word
$\mathrm{D}(n)$	900000000000000	unsigned word
$\mathrm{D}^2(n)$	22050000000000000	unsigned word
$\mathrm{D}^3(n)$	595350000000000000	unsigned word
$\mathrm{D}^4(n)$	17562825000000000000	unsigned word; last factored operand
$\mathrm{D}^5(n)$	553228987500000000000	unsigned 128-bit
$\lfloor\sqrt{Q_5(n)}\rfloor$	553508012914801786476	unsigned 128-bit

Table 4: Conservative representation bounds, not observed coordinate maxima. The supported cap $4 \cdot 10^{13}$ is distinct from the completed census endpoint 30429212500.

which requires 138 bits. Thus even an unsigned 128-bit sum is not sufficient at the supported cap; GMP is necessary for this implementation's full domain. The last factored operand fits below 2^{64}, so no larger-integer factorization theorem is assumed. Word modular products use 128-bit intermediates. Prime multiplicity is at most 63 for these word operands, within the 64-slot factor array. Decimal conversion, allocation growth, loop endpoints, and narrowing conversions are checked. These bounds establish representation capacity; they do not assert the practical feasibility of a complete census to the cap.

5.3 Parallel work and completed-block recovery

The first-derivative sieve is serial within a block. The remaining per-input work uses OpenMP dynamic scheduling in chunks of 64; each input owns one result flag. After the join, an incomplete flag prevents the block from being committed. Positive rows are replayed and appended in input order serially. The frontier advances only after every input in the block has completed; an interrupted block and its tentative hits are discarded and recomputed on resumption. There is no resumption inside a single factorization or input.

For N_I inputs and block size β, let $K = \lceil N_I/\beta \rceil$. If G is the maximum block count per operational segment, set $\sigma = \lceil K/G \rceil$. Segment j, for $0 \le j < \sigma$, owns block indices

$$\lfloor jK/\sigma \rfloor, \ldots, \lfloor (j+1)K/\sigma \rfloor - 1.$$

These intervals are disjoint and cover all K blocks; adjacent endpoints coincide, and their block counts differ by at most one. Only the final input block is clipped to the configured endpoint. The campaign used $\beta = 262144$, eight workers, and four immutable configurations.

Persistent states use temporary-file replacement with file and directory synchronization, internal digests, and a separate seal for a complete segment. Immutable configuration and source/build/executable identity checks bind the run records to the executed program. These checks detect accidental corruption and inconsistent coverage, rather than protect against a trusted owner rewriting all evidence. Sealed segment states are authoritative; aggregate tables are checked against them. A successful single segment is not completion of the whole configured target. Mathematical coverage requires every configured segment to be complete, not merely an exit status or the largest input assigned to a worker.

6 Computational result and validation

Theorem 6.1 (Computational extension, with explicit dependencies)**.** *Accept the indexed ten-term prefix in [1]. Subject to the correct execution of the reviewed exact kernel of Section 5, including its stated finite-domain primality premise, and the completed segment evidence in*

i	$a(i)$	$\mathrm{D}(a(i))$	$\mathrm{D}^2(a(i))$	$\mathrm{D}^3(a(i))$	$\mathrm{D}^4(a(i))$	$\mathrm{D}^5(a(i))$	R
1	4031	168	332	336	832	2560	4873
2	10823	216	540	1188	2484	5076	12281
3	416959	24544	63664	130464	457488	1526256	1653567
4	496939	7484	7488	28032	107840	345408	615453
5	1354980	2201476	2201480	3791236	3816012	5088028	8145700
6	9146115	14859963	14859990	25590843	25758081	34344189	54983475
7	38949392	81214608	190342672	409971536	859225680	2463114336	2649079248
8	44472866	22258485	19398957	6466322	3262101	1100831	53881926
9	262908396	548198604	1284813036	2767307868	5799773340	16626021768	17881284924
10	380264131	29251100	41359540	49631468	49948220	67928164	395903259
11	1058578125	1719903125	1719906250	2961903125	2981259375	3975021875	6363828125
12	1072281223	85244735	27198937	1431542	920291	1	1076009463
13	1774631673	3700340577	8672487993	18679328109	39148470045	112225646934	120698673237
14	2539872501	928555631	79073578	39536791	5891811	1986491	2705738777
15	9873543431	747786611	127012442	64347665	15088563	12624681	9902863451
16	30429212500	63448912500	148705212500	320290262500	671270062500	1924308075000	2069593162500

Table 5: The complete locally consolidated sixteen-term prefix and all ordered witnesses. Each row has five derivative transitions and satisfies $a(i)^2 + \mathrm{D}(a(i))^2 + \cdots + \mathrm{D}^5(a(i))^2 = R^2$. The horizontal break separates the inherited ten-term baseline from the six additional indexed terms. No coordinate is sorted or suppressed.

Stage	Inclusive input interval	Segments	Positives	Scan seconds
1	380264132 . . . 1058578125	1	1	584.88
2	1058578126 . . . 1774631673	1	2	666.78
3	1774631674 . . . 10000000000	13	2	9243.99
4	10000000001 . . . 30429212500	35	1	26457.62
Total	380264132 . . . 30429212500	50	6	36953.27

Table 6: Completed consecutive campaign intervals. Times are per-stage totals of the 50 recorded GNU time wall measurements of scanning invocations, one per segment, shown at their recorded precision. They exclude pauses, subsequent checks, and other work. Environment and timing scope are stated in Section 6.1.

Table 6, the only members of S_5 in the inclusive interval $[380264132, 30429212500]$ *are*

$$1058578125, \quad 1072281223, \quad 1774631673,$$
$$2539872501, \quad 9873543431, \quad 30429212500.$$

They are respectively $a(11), \ldots, a(16)$*. Their positive witnesses are independently reconstructed in Table 5; completeness of the new negative census depends on the production execution and seals, and has not been independently repeated over the full interval.*

Computational proof. The sieve and exact factor/square kernel test the defining predicate at every completed input, as described in Section 5. All 50 segment states are complete, with the frontier equal to their inclusive high endpoint, and agree with their seals and aggregate hit records. Their within-stage partitions and cross-stage joins have neither gaps nor overlaps. The union is exactly the interval in the statement. Its cardinality is

$$30429212500 - 380264132 + 1 = 30048948369.$$

The ordered union of the retained hits contains exactly the six displayed values, leaving $30048948369 - 6 = 30048948363$ negatives in the production census. The independently checked witnesses prove their membership. Given the accepted baseline and the production negative evidence, the six increasing hits receive exactly indices 11–16. File-integrity and positive checks by themselves would not prove the exclusion claim. □

Corollary 6.2. *Under the computational dependencies of Theorem 6.1,*

$$30429212500 < a(17) \leq 205397184375. \tag{18}$$

Proof. The strict lower bound follows from complete coverage through $a(16)$. The family F_2 at $(p, q) = (3, 5)$ gives $2434337 \cdot 3^3 \cdot 5^5 = 205397184375$, with witness

$$(205397184375, 428280159375, 1003760184375,$$
$$2161959271875, 4531072921875, 12989079506250,$$
$$13969753846875).$$

Its five derivative identities and square identity were also checked directly. It is greater than the completed endpoint, so a next term exists no later than this positive. No coverage of the intervening gap, or index for this upper witness, is asserted. □

6.1 What the validation establishes

The production run used an AMD Ryzen 9 7940HS with eight physical cores and eight campaign workers, Linux Mint 22.3 with kernel 6.17.0-42, GCC 13.3.0, GMP 6.3.0, and OpenSSL 3.0.13. The frozen build uses `-O3 -std=c17 -fopenmp`, strict warnings promoted to errors, and links GMP and libcrypto; no processor-specific tuning flags were added. The recorded scan time sums 50 GNU time measurements, one per completed segment. Table 6 aggregates them by stage; the total is approximately 36953 seconds, about 10 hours 16 minutes. It is active scan wall time, not elapsed calendar time; native monotonic nanoseconds have a slightly different measurement scope. Maximum recorded campaign resident memory was 10656 KiB. This measured RSS is distinct from conservative allocation bounds and from earlier diagnostic RSS measurements.

The archived independent Python checker uses exhaustive trial division and `math.isqrt`, without importing the C sieve, Miller–Rabin, rho, or GMP arithmetic. Each stage invocation checks the ten baseline positives and that stage's retained hits. A separate finite arithmetic script, `finite_audit.py`, reconstructs all sixteen full witnesses: 80 derivative identities and 16 square identities. It also checks all normalized transitions, both seeds, the unindexed upper

witness, and 48 extra transitions through depth 8, three for each of the sixteen inputs. Table 3 displays the 27 continuations for the nine singleton witnesses. These are finite positive arithmetic and structural checks, not a second global census.

Independent trial-division enumeration was additionally recorded for one 4096-input window at each end of each of the four stages, giving eight disjoint windows and 32768 distinct inputs. They contain three positives and 32765 negatives. The exact intervals are

$$\begin{aligned} [380264132, 380268227], \quad & [1058574030, 1058578125], \\ [1058578126, 1058582221], \quad & [1774627578, 1774631673], \\ [1774631674, 1774635769], \quad & [9999995905, 10000000000], \\ [10000000001, 10000004096], \quad & [30429208405, 30429212500]. \end{aligned} \tag{19}$$

Separate development windows contain 5098 distinct inputs, with four positives and 5094 negatives. That count is not added to the post-run window count: the development and campaign sets have overlaps.

The retained development matrix has sixteen obligation groups. Strict GCC and Clang 18.1.3 builds, independent small windows and positive fixtures, one/four/eight-worker agreement, balanced partitions, and signal/recovery checks exercise different contracts. Numerical tests include hard factor operands, forced exact fallback, divisors of the fixed primality bases, cap/refusal boundaries, and genuine wide paths: one input produces a 68-bit fifth derivative and a 136-bit sum. Synthetic wide root/elapsed records test serialization and are not presented as mathematical witnesses. Corrupt or forged records, identity mismatches, locking, malformed configuration, and injected synchronization/rename/write failures test fail-closed persistence behavior.

ASan/UBSan checks passed in their recorded bounded scopes with LeakSanitizer disabled because of its launch-environment limitation. Qualified OpenMP ThreadSanitizer checks used Clang/libomp/Archer, a silent synchronized control and a deliberately racy control that was detected, before testing the candidate at one, four, and eight workers. Static analysis records include cppcheck and clang-tidy. These tests supplement the numerical and ownership arguments; neither sanitizer silence nor thread agreement is a universal proof of absence of defects. External code reports were adjudicated against the source and these records; their verdicts do not replace arithmetic evidence or constitute review of this manuscript.

6.2 Reproducing a record audit without changing the baseline

The compact publication package retains both canonical tables, the original checker, and the complete sealed records of all four stages. From its root,

```
python3 -B check/verify.py
python3 -B check/verify.py windows
python3 -B check/verify.py native
```

The default command verifies the exact payload inventory, archive members, record seals and stage joins; it regenerates both canonical tables and compares the finite arithmetic output. The `windows` mode additionally enumerates the recorded independent windows. The `native` mode builds and tests the public source in temporary storage and uses the exact original executable to audit all four historical stages. These finite checks do not repeat the global negative census.

The historical checker deliberately binds its ten-term baseline by identity; the consolidated b-file has sixteen rows. The driver restores the original reference layout in temporary storage, without changing that baseline or any historical checker constant. Its historical `UPPER` constant denotes $1058578125 = a(11)$, not the present bound 205397184375. The lossless archives use compact member names and contain explicit mappings, byte sizes and hashes for restoring original paths. The original source, build file and executed binary retain their historical identities.

The public build file changes only the two `check` command paths to use this driver; its compiler flags and the C source are unchanged. Its new identity and any new executable are distinct from the historical run identities. Explicit Python failure checks remain active with optimization enabled. The README gives dependencies and optional restoration instructions. No PDF is required to perform these checks.

The two canonical tables have distinct roles. The b-file lists consecutive index–value pairs through index 16. The documentary a-file contains the eight columns of Table 5 and makes every positive witness directly checkable. It is not a sparse list of later indexed terms beyond an unresolved gap. The public repository and persistent archive are cited in [17]; their roles are specified in Data availability below.

7 Related sequences, literature, and open questions

OEIS A003415 is the arithmetic derivative itself [12]. The lower-depth square-sum entries A230543, A249105, and A249106 use depths $2, 3, 4$ respectively, while A249110 uses depth 8 [13]. A210503 concerns primitive Pythagorean triples at depth 1 [14]; its primitivity condition is additional to a square-sum test. Thus the family constants, positivity policies, widths, term indices, and validation domains from these neighboring problems cannot be transferred to A249107. Only the underlying product-rule reasoning applies after the relevant transitions have been checked.

The persistence in Proposition 4.3 is an unconditional statement for the explicit S_h convention of this paper. In particular, it supplies a zero-extended depth-eight witness with the same root. The current A249110 entry's displayed Maple factorization expression does not explicitly implement a separate $\mathrm{D}(0) = 0$ branch. We have not executed that Maple program on a zero tail or assigned N an index in that entry. Identification with an implemented external predicate must therefore retain the zero-convention qualification. The original A249107 comment about no eight- or nine-tuple examples up to 10^8 has that different finite domain and does not contradict $N > 10^9$.

The entries A258644 and A258645 give fourth and fifth arithmetic derivatives as functions of their input index; A258651 is the array $A(n, k) = \mathrm{D}^k(n)$, flattened by antidiagonals [15]. Their values encode coordinates of a chain. A square identity and a rank in A249107 do not follow from being a value in one of these arrays. The array's explicit zero and unit conventions agree with those used here.

The theoretical distinction relevant to the scale classification is the domain of the logarithmic equation. Ufnarovski and Åhlander study rational solutions, while Kovič treats positive integer solutions at integer rates [2, 3]. The complete integer-scale statement here adds the residue-core nonnegativity condition to the squarefree denominator requirement, following the author's fixed-form work [5, 6]. Deza and coauthors [7] examine arithmetic constants and rational differential equations, including multiplicative relations between solutions. These rational constructions are relevant background, but do not replace the integrality, all-five- transition, and square-identity conditions for a depth-five witness. The cited works are used for these specific contributions; no longer ranked A249107 dataset was admitted from them.

Several questions remain open within this study. What is $a(17)$ inside the interval (18)? Are there infinitely many distinct admissible primitive ordered forms at depth five, or infinitely many singleton forms? Does an admissible primitive form of this sequence have a noninteger rate? The observed eleven forms have rates only $0, 1, 2$, but this finite sample and the fixed-form theorem supply no answer for unobserved forms. Likewise, whether there are infinitely many terminating members of S_5 is separate from the three terminating examples proved here. No density law or global conjecture is proposed on the strength of sixteen terms. The existence and infinitude of the two displayed classes, in contrast, are theorems and require no such extrapolation.

8 Conclusion

Complete added interval coverage assigns six consecutive terms beyond the accepted ten-term prefix, ending at $a(16) = 30429212500$, with the stated production dependence of the negative census. The positive witnesses, their primitive forms, and their scale transitions admit short exact checks independent of that census. Three additional forms occur among the new terms, while three of the added indices belong to families whose positive values were already available before enumeration. The resulting eleven observed forms separate two infinite homothetic classes from nine singleton classes without exhausting the possible forms of the sequence. The finite trajectory decision also separates homothetic structure from dynamics: both divergence and eventual zero occur among singleton forms. The persistent identity at $a(12)$ illustrates why the convention for later zero coordinates matters in comparisons across depths. The next rank remains bounded by a proved family witness, and its exact value requires additional ordering evidence.

Data availability

The code, immutable configurations, sealed segment records, canonical b-file, documentary witness table, independent checker and frozen reference inputs, finite manuscript-arithmetic script and outputs, historical environment captures, and validation summaries are publicly available in the GitHub repository and Zenodo archive cited in [17]. The cited Zenodo DOI identifies the release series; each deposited release has its own version-specific DOI. The repository includes this monolithic source and its corresponding PDF, under the MIT license. The lossless evidence archives and portable checks are described in Section 6.2. The mathematical tables and proofs needed to read the results are contained in the source and PDF.

AI use disclosure

OpenAI Codex assisted with the project analysis, implementation workflow, local artifact checks, and preparation of this manuscript. The supplied code review material explicitly reports contributions from Anthropic Claude and the GLM family; other exact backend identities and versions are not uniformly documented. The supplied manuscript-review reports identify OpenAI ChatGPT, Anthropic Claude, GLM, MiniMax, and Qwen; several other reports give no model identity. These are reported identities, not independently authenticated backend versions. These reported tools are not coauthors, in accordance with the responsibility and transparency principles of [16].

Mathematical analysis. AI assistance supported the reconstruction of the sequence definition from OEIS, the product-rule and scale proofs, normalization of the full ordered witnesses, and investigation of terminating and divergent trajectories. The mathematical statements are supported by the displayed proofs, primary source readings, exact witness tables, and separately executed finite integer calculations.

Algorithm and code. Codex assisted with the specialized C implementation and its numerical, parallel, and recovery contracts. The supplied Claude and GLM contributions include external code review; they do not establish independent enumeration. The final computational claims are tied to the executed program, immutable run records, seals, known-term replay, independent trial-division checker, and scoped sanitizer and fault-test evidence. Unverified execution statements in review prose were not substituted for local validation records.

Manuscript. Codex assisted in drafting English prose and LaTeX, checking literature metadata and contributor credit, and internally reviewing claim scope, chronology, and availability. This manuscript incorporates pertinent findings from the supplied external AI manuscript reviews after adjudication against the source and retained evidence. The reports' execution statements have not been added to the local validation results. Final numerical claims rely on local artifacts and validation, not conversational assertions. Carlo Corti retains full scientific responsibility for the mathematical statements, algorithm, implementation, numerical results, validation boundaries, and conclusions.

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